

ENTERPRISE HOME LAB

Oracle VirtualBox Enterprise Lab Setup

Installation, network segmentation, appliance import, validation, and
troubleshooting

PORTFOLIO DOCUMENTATION

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Executive Summary

This project documents the deployment of Oracle VirtualBox 7.2.12 on a dedicated Windows 11 laboratory account. The environment was built to support repeatable cybersecurity training while keeping lab files, virtual disks, and network settings separated from the personal Windows profile.

Project Outcome

VirtualBox and the matching Extension Pack were installed, two isolated NAT networks were created, the NGT-provided Kali appliance was imported, and the final VM was validated with dual virtual network adapters.

Component	Final Configuration
Host	Windows 11 on GEEKOM A8 Max
Hypervisor	Oracle VirtualBox 7.2.12
Extension Pack	7.2.12r174389
Lab account	KevinLab
Primary VM	Kali Linux - NGT
VM resources	8 GB RAM, 4 vCPUs, 80 GB VDI
Networks	NCSA-LABS 10.0.2.0/24 and HOME-LABS 10.0.20.0/24
Security boundary	NAT networks with DHCP; lab-specific account and storage path

Host and Account Architecture

- VirtualBox was installed with administrator approval because hypervisor networking drivers require elevated permissions.
- The application and VM files were operated from the dedicated KevinLab Windows profile.
- The default VM path was kept under C:\Users\KevinLab\VirtualBox VMs.
- Personal files and credentials were not stored inside the lab virtual machines.

VirtualBox Installation

1. Launch the Oracle VirtualBox 7.2.12 installer.
2. Review the default application, USB, networking, and Python support features.
3. Keep the default installation path unless the host storage plan requires another location.
4. Accept the temporary network interruption warning.
5. Complete installation and launch VirtualBox Manager.

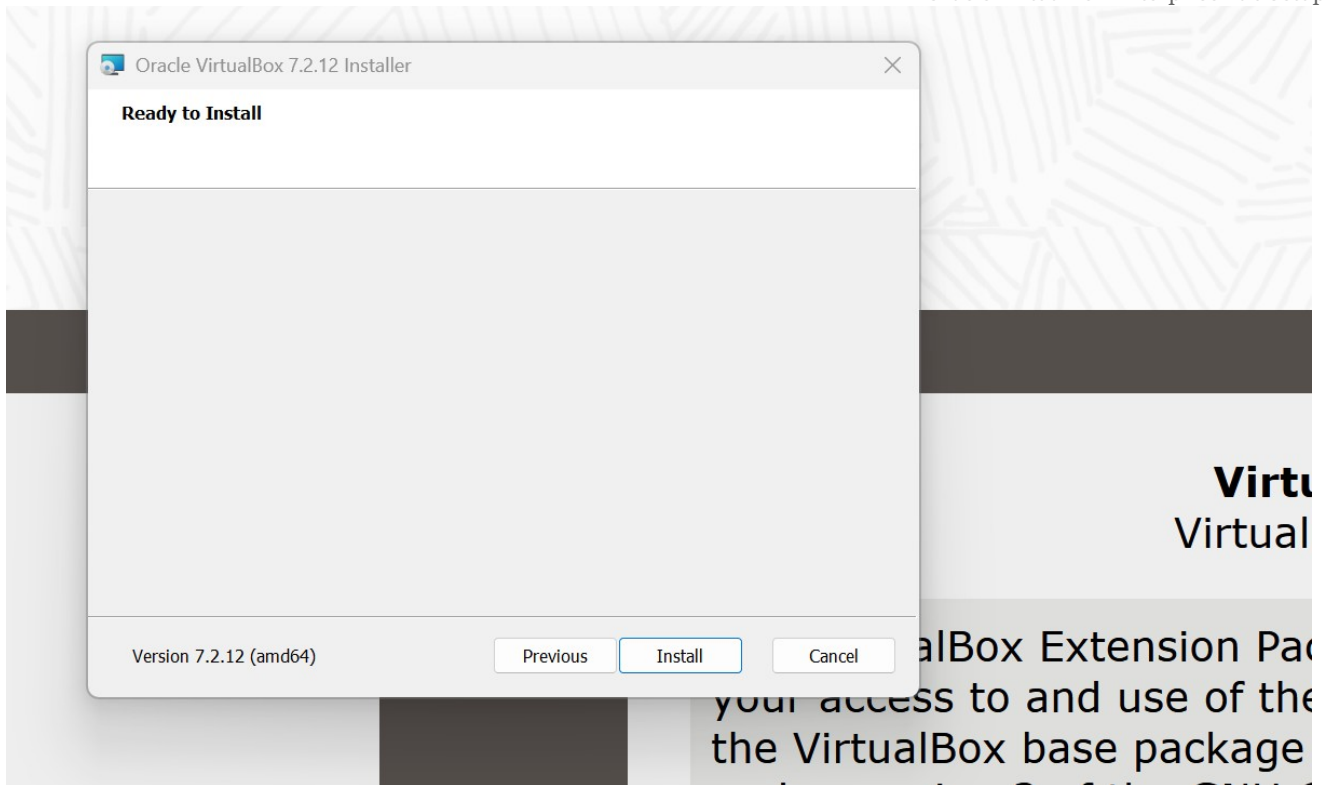


Figure 1. Installer welcome screen

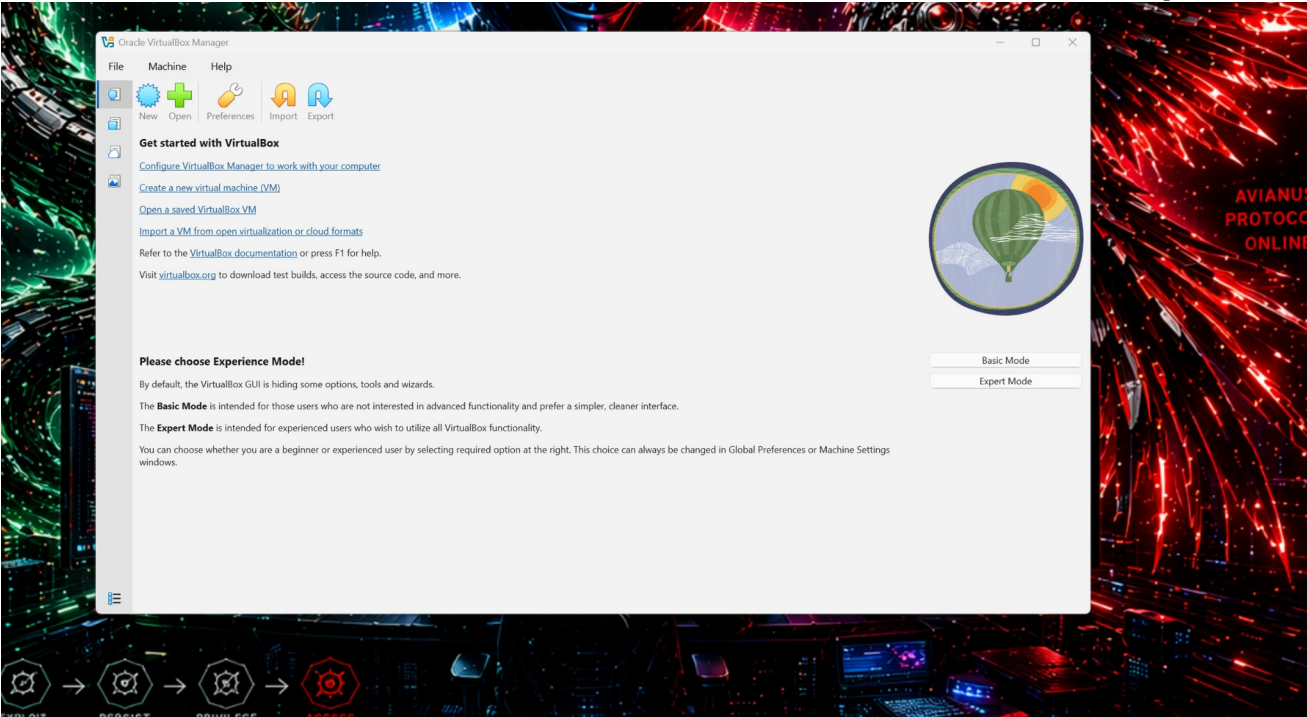


Figure 2. Custom setup features and installation path

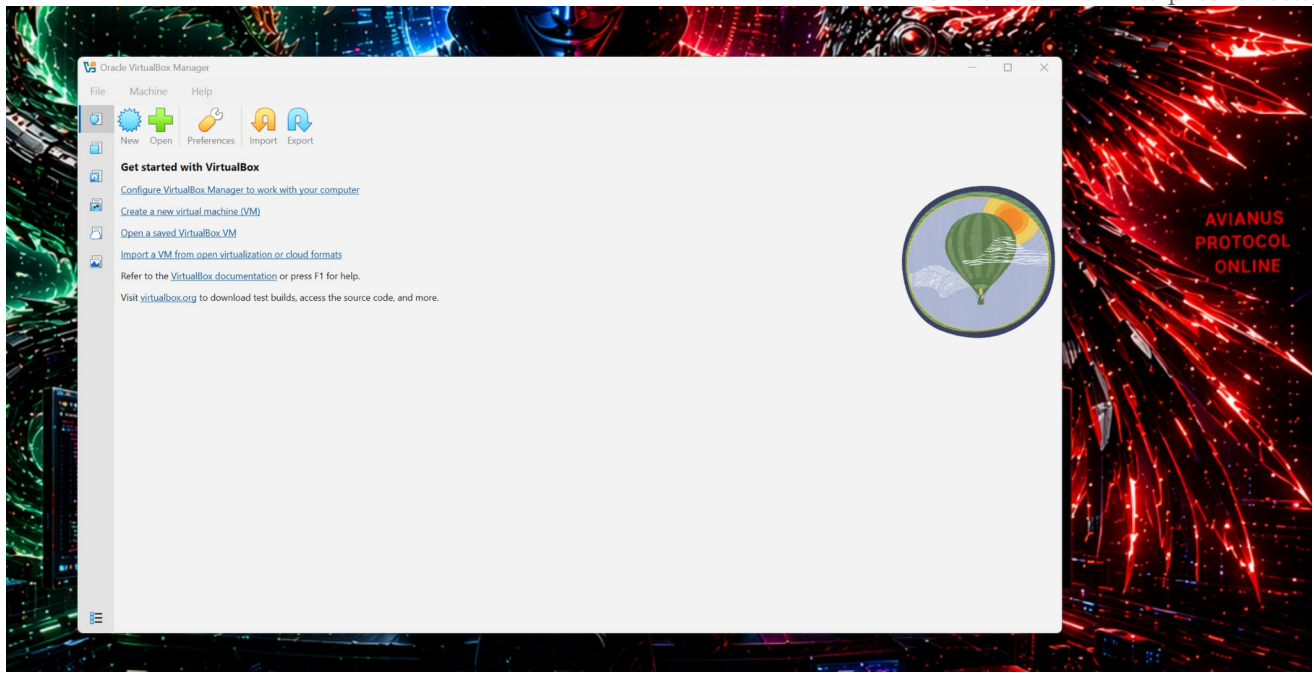


Figure 3. Startup integration options

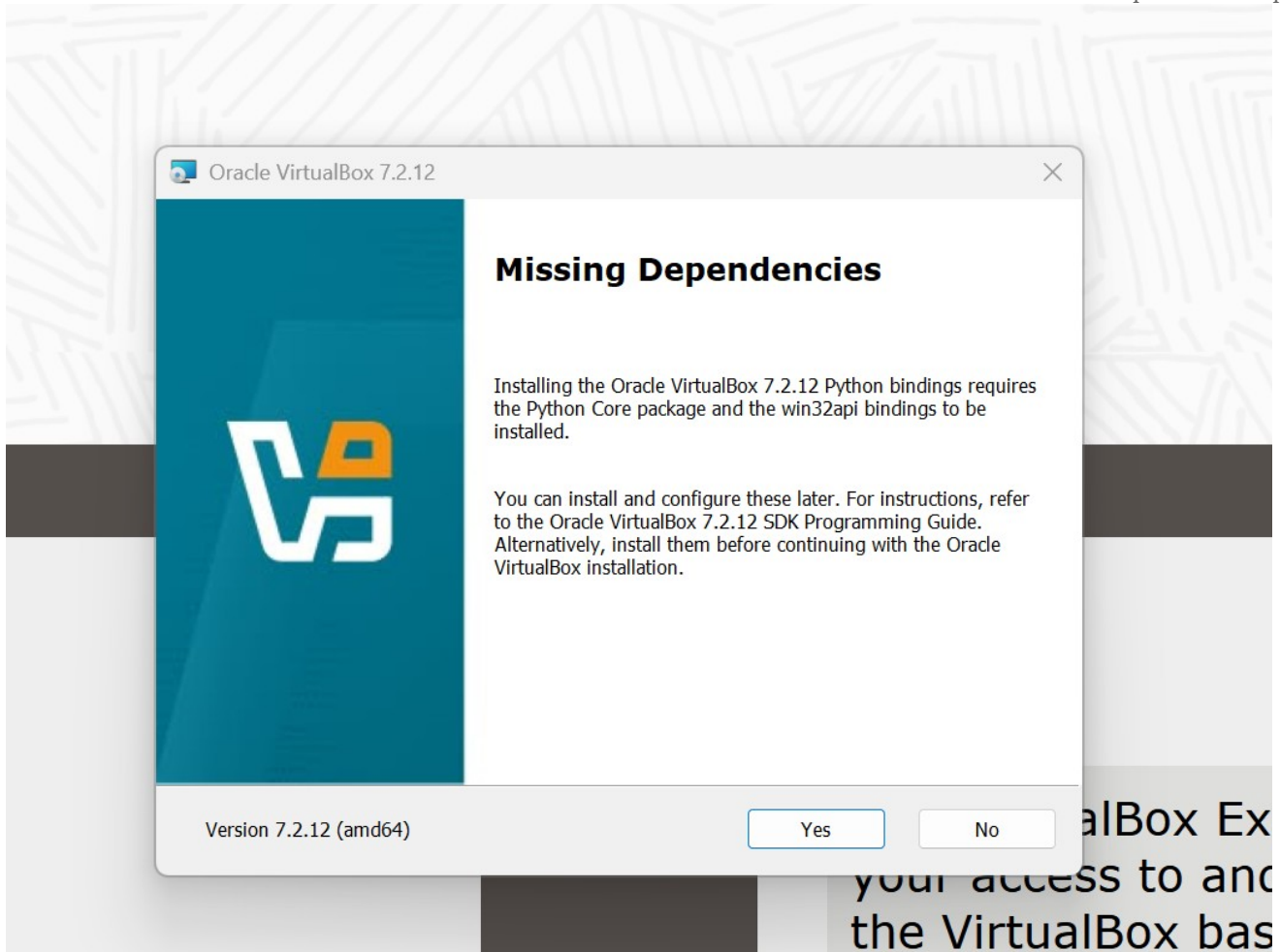


Figure 4. Missing Python dependency notice

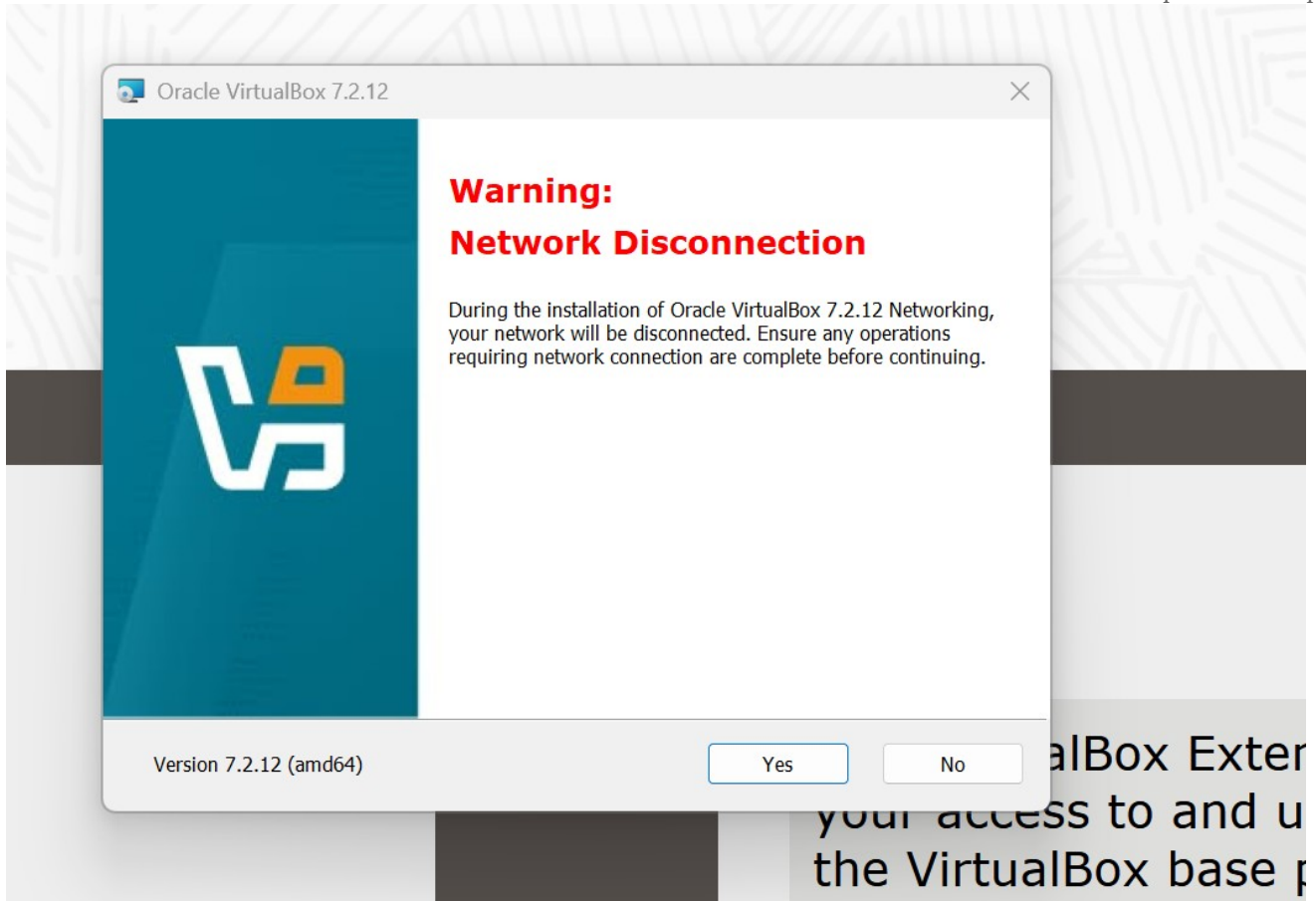


Figure 5. Network disconnection warning

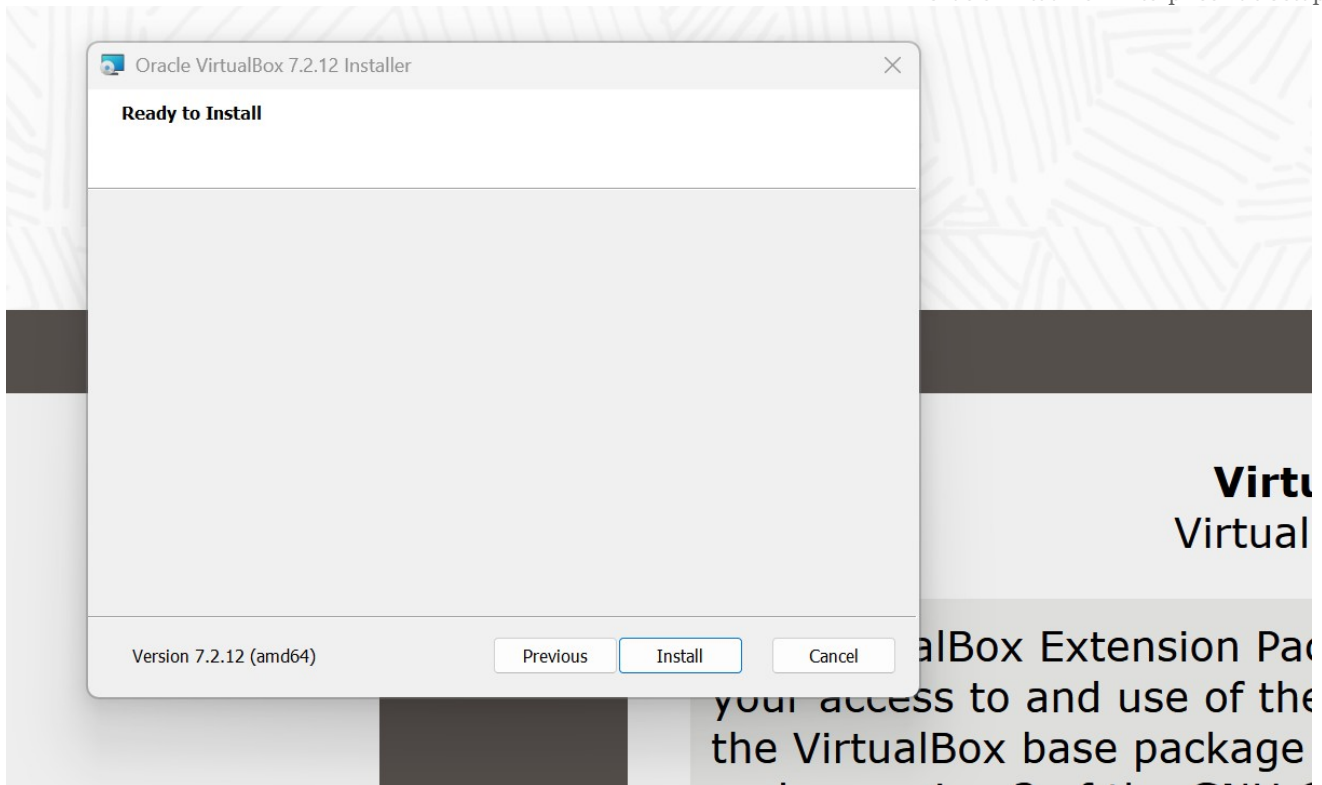


Figure 6. Ready to install

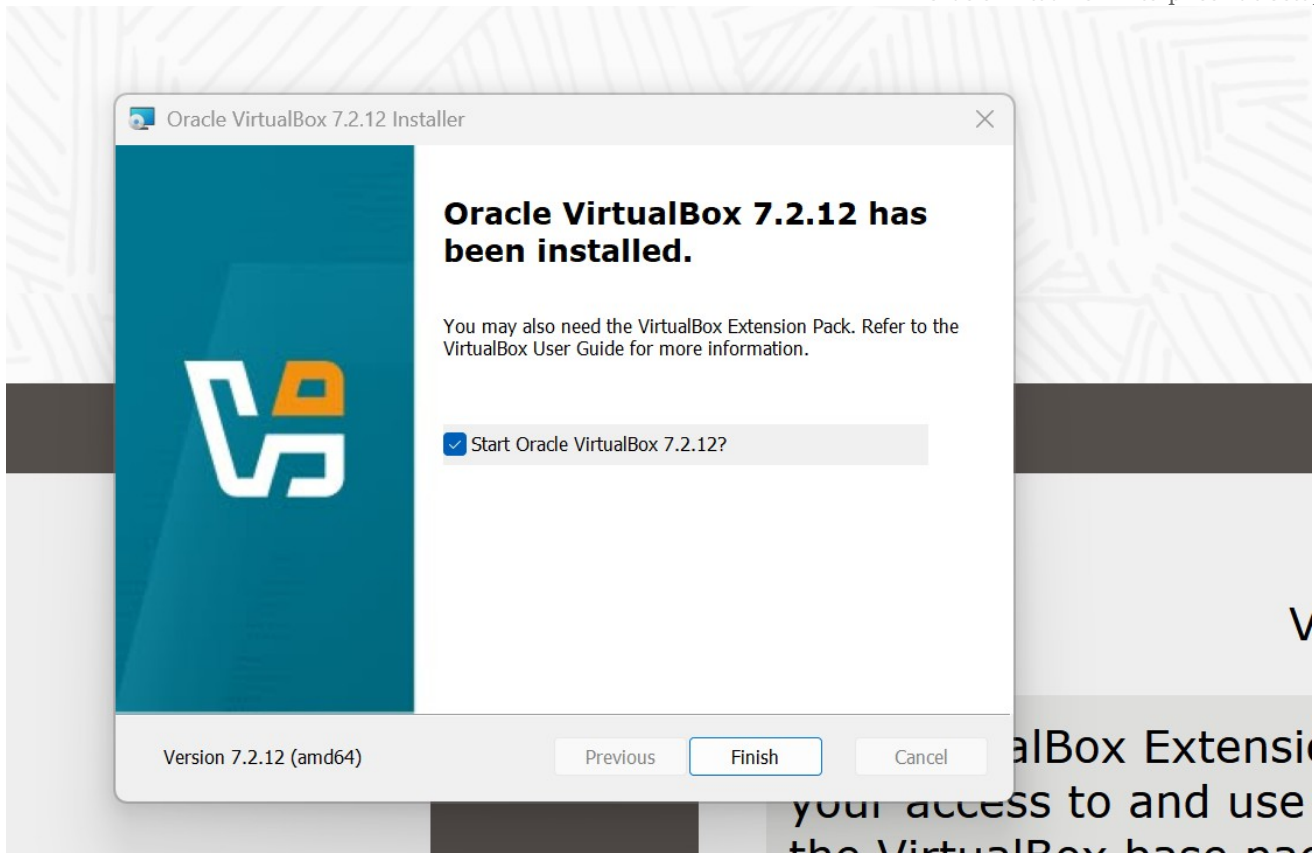


Figure 7. Installation completed

Extension Pack Installation

The matching Extension Pack was installed to provide additional USB, remote-display, and integration features. The first attempt failed because the helper process required membership in the local Administrators group. Re-running VirtualBox with administrator approval resolved the issue.

Troubleshooting Lesson

A matching product version is necessary, but the Extension Pack also requires elevated installation privileges on Windows. The error was documented, corrected, and then validated in VirtualBox Manager.



Figure 8. Extension Pack confirmation

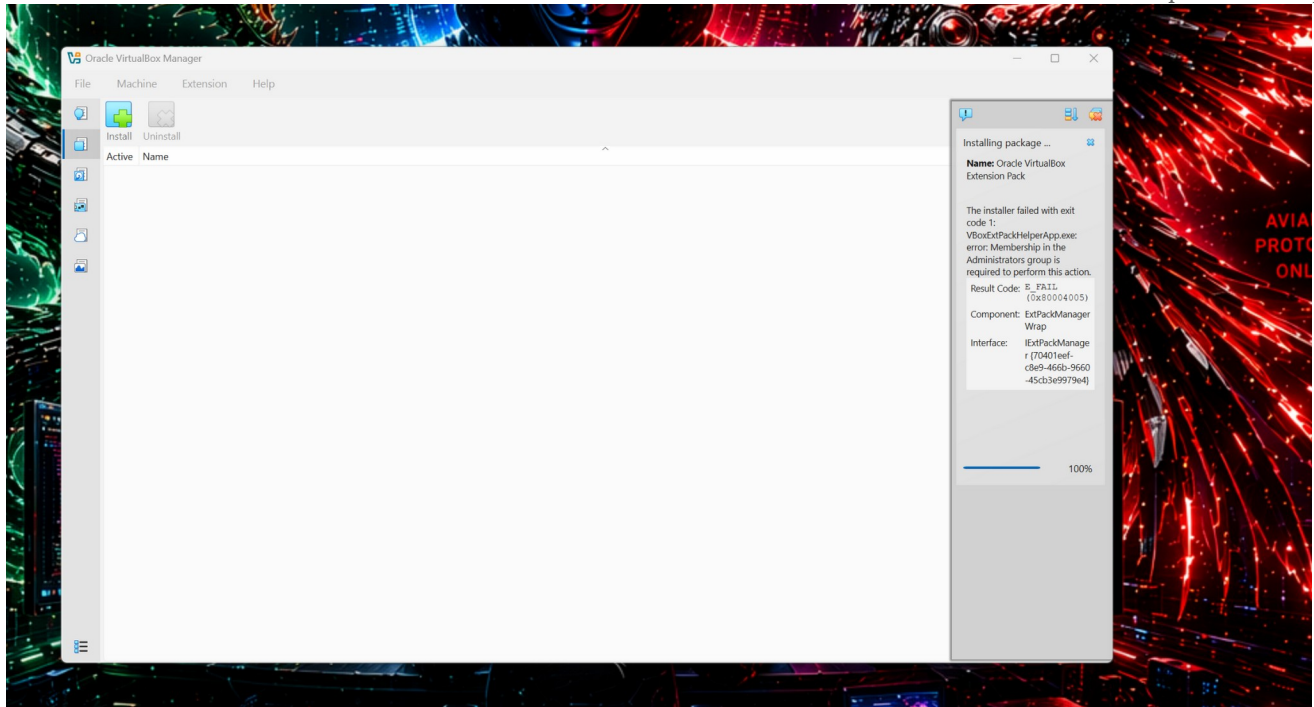


Figure 9. Extension Pack administrative permission error

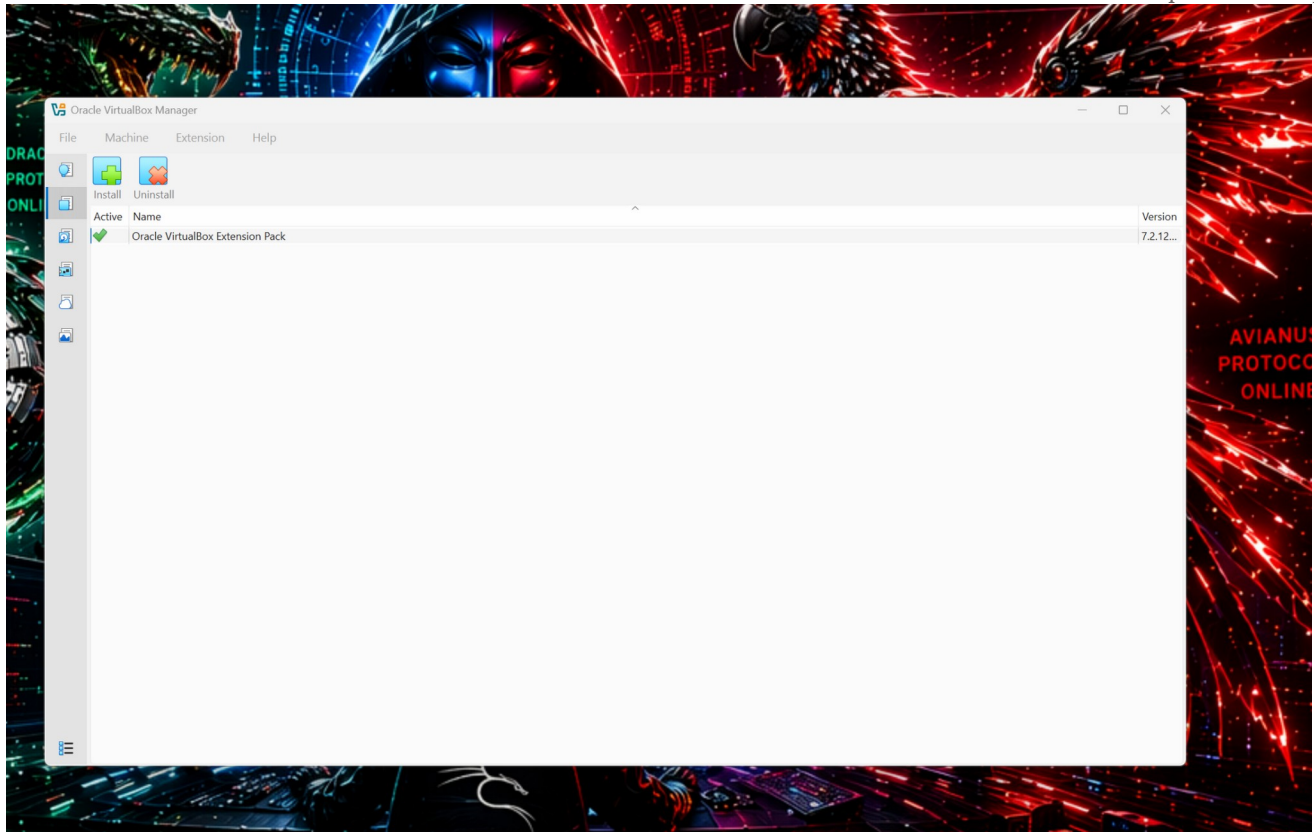


Figure 10. Extension Pack successfully installed

Virtual Network Design

Two NAT networks were created to provide controlled address spaces for separate training contexts. Both networks use VirtualBox DHCP and remain behind the host network address translation boundary.

Network	IPv4 Prefix	Purpose
NCSA-LABS	10.0.2.0/24	Course-aligned lab network
HOME-LABS	10.0.20.0/24	Independent home-lab exercises

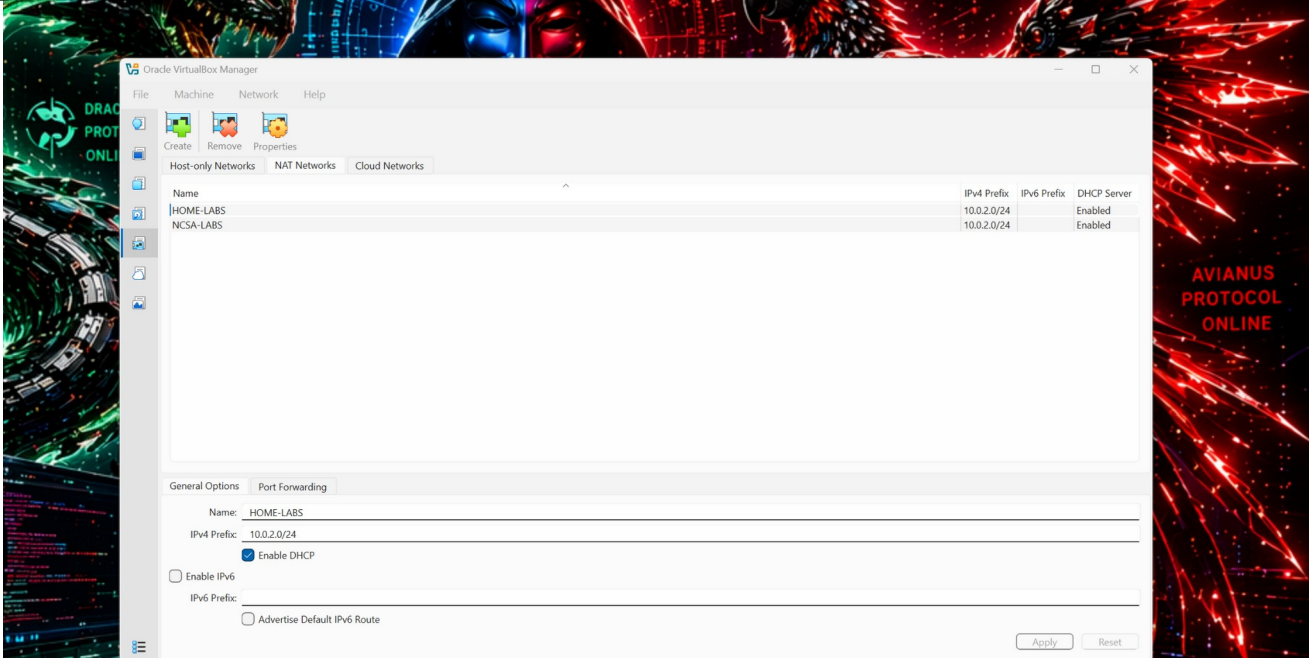


Figure 11. NAT networks HOME-LABS and NCSA-LABS



Figure 12. Adapter 1 connected to NCSA-LABS

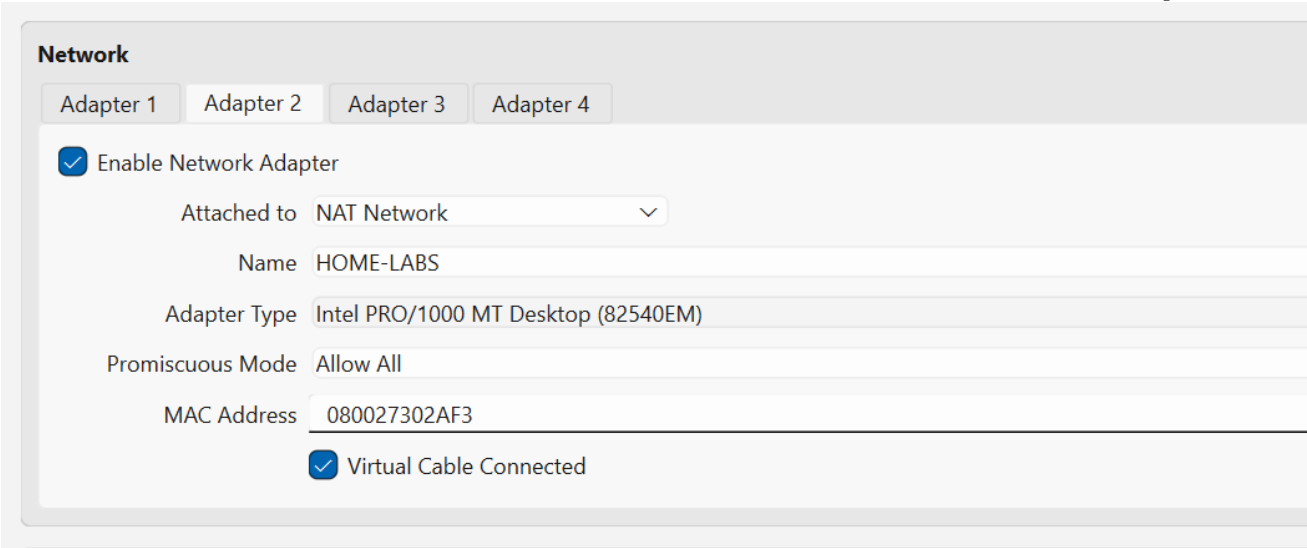


Figure 13. Adapter 2 connected to HOME-LABS

Kali Appliance Import

6. Select Import from the VirtualBox Manager.
7. Choose the NGT-provided Kali Linux OVA.
8. Review guest OS type, CPU, RAM, storage, USB, and sound settings.
9. Select a machine base folder within the intended Windows account.
10. Import the hard drives as VDI and complete the appliance import.

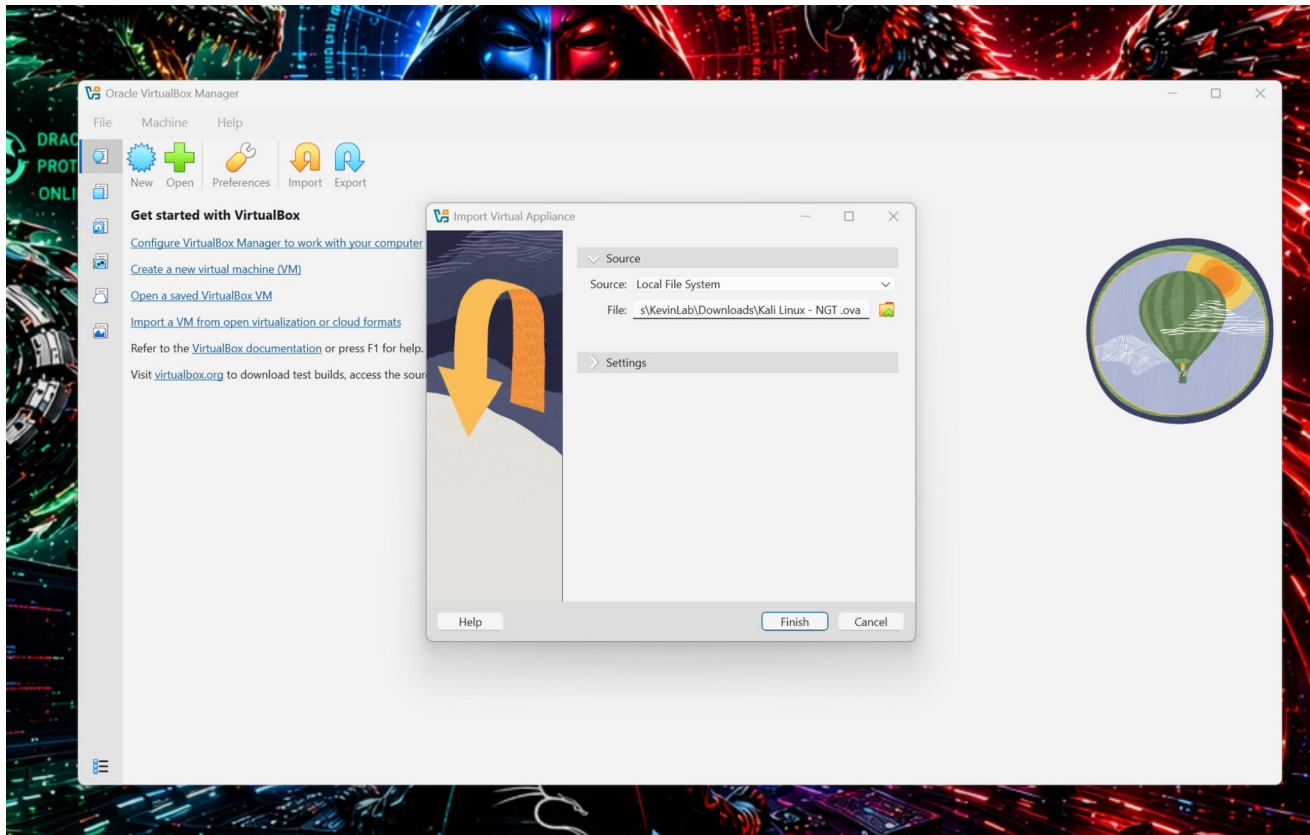


Figure 14. Kali appliance selected for import

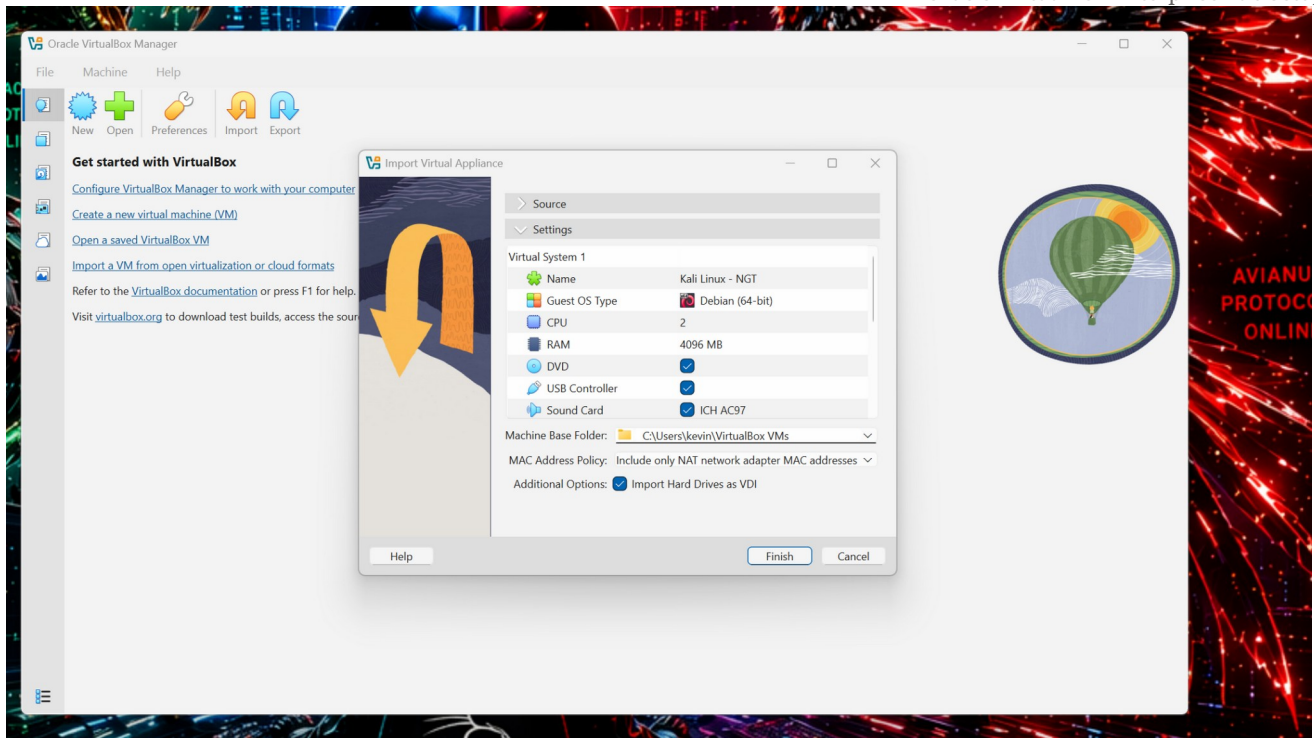


Figure 15. Import settings summary

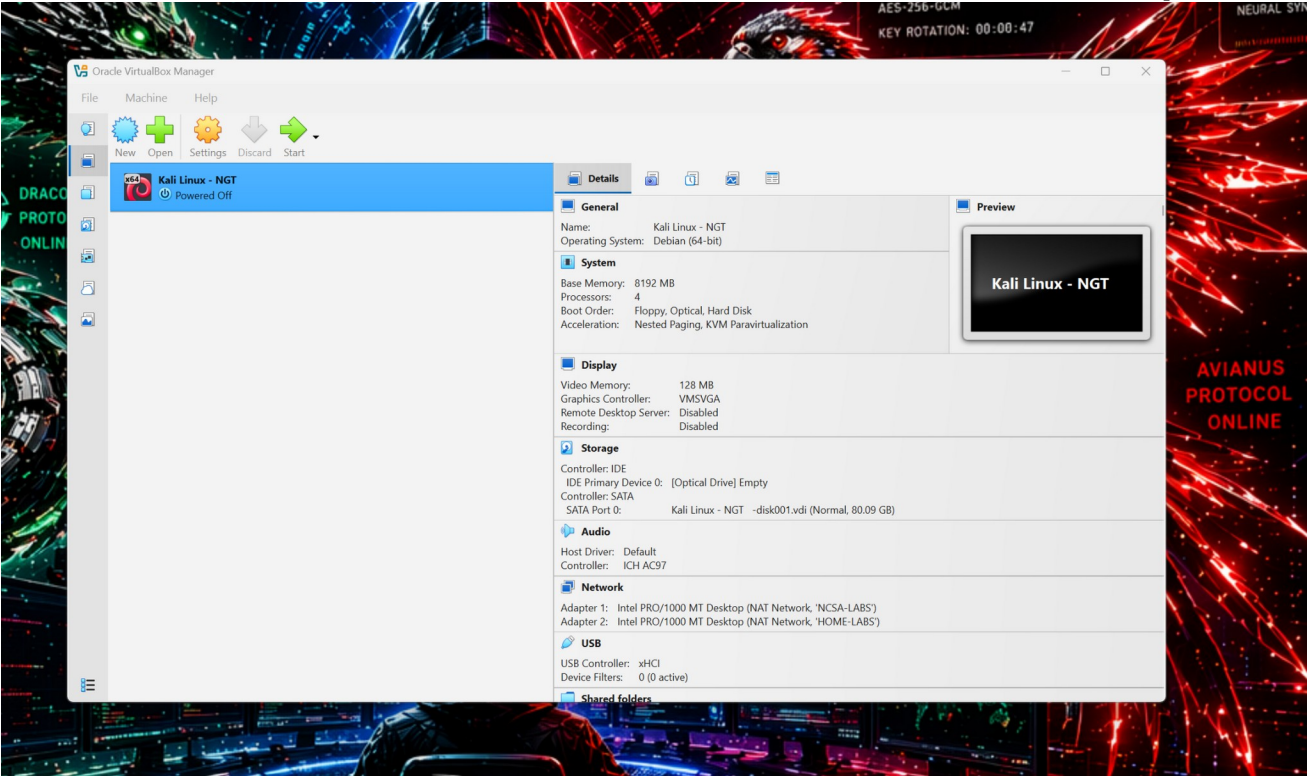


Figure 16. Final Kali VM resource and network summary

Validation and Troubleshooting

During recovery from duplicate or misplaced VM folders, the .vbox configuration and .vdi virtual disks were located, reviewed, and re-associated with the correct lab account. This demonstrated practical understanding of VirtualBox machine registration and storage dependencies.

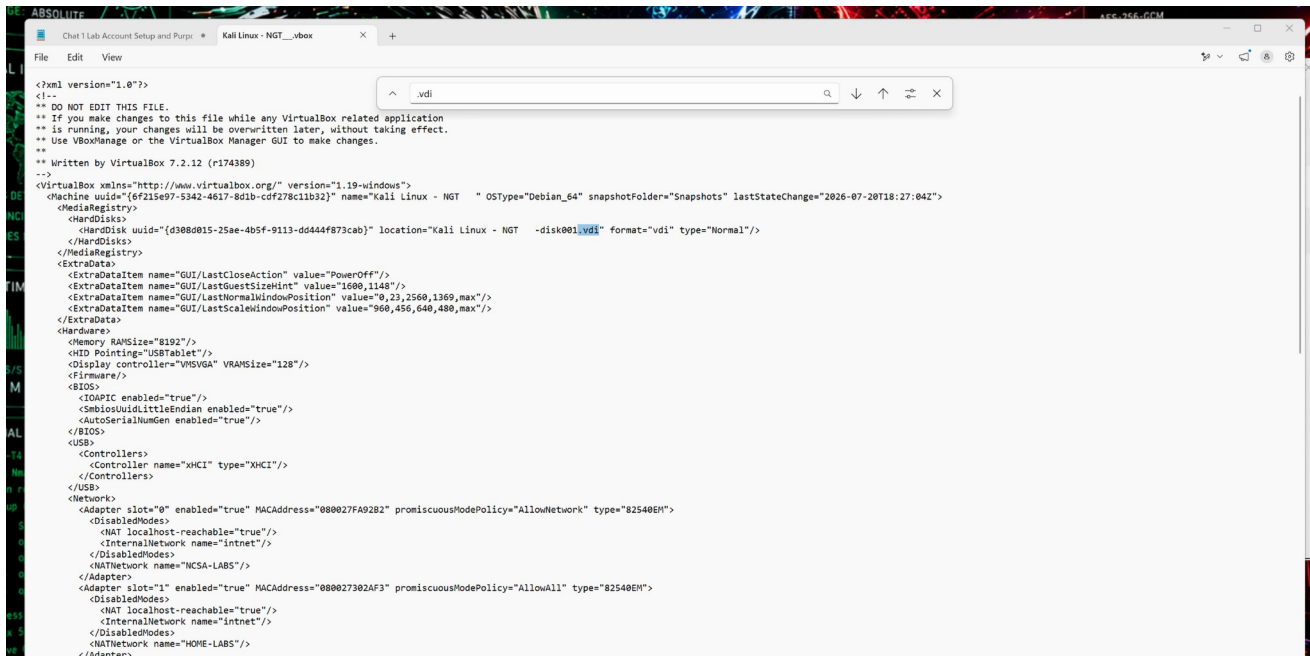


Figure 17. Virtual disk files located on the lab account

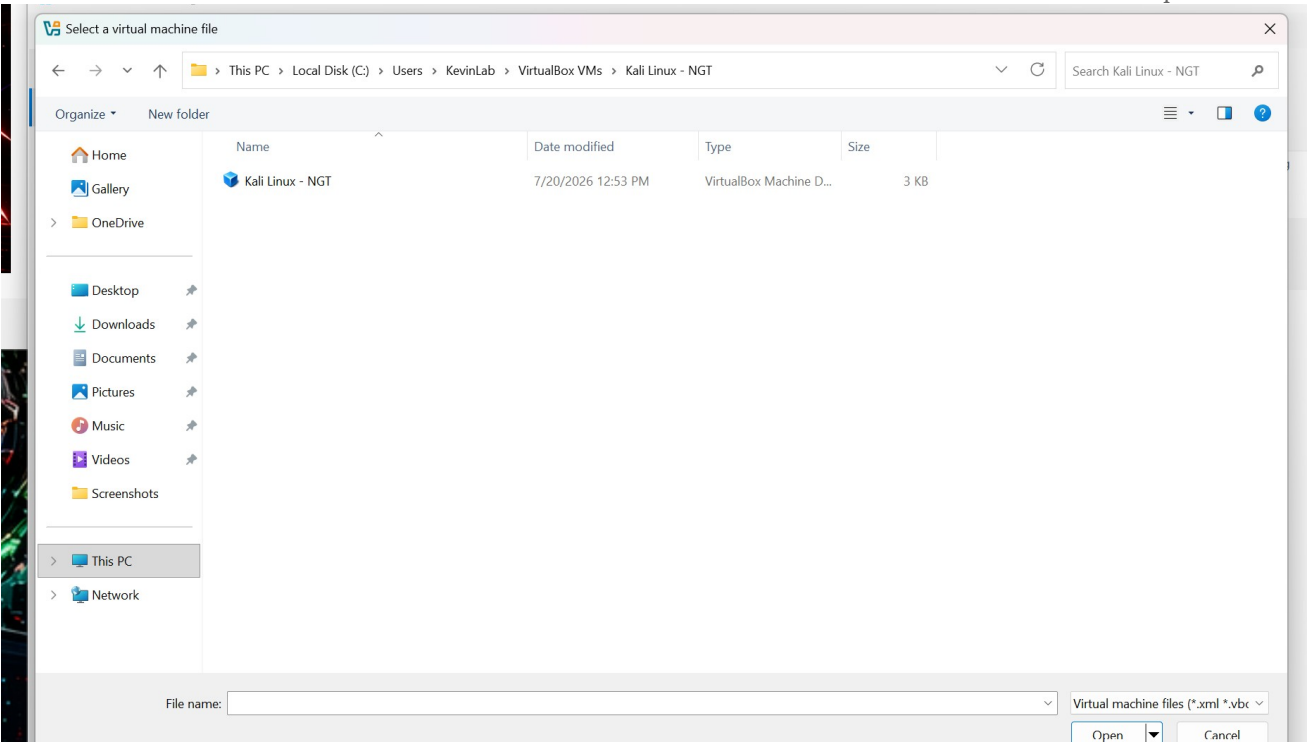


Figure 18. VirtualBox configuration file validation

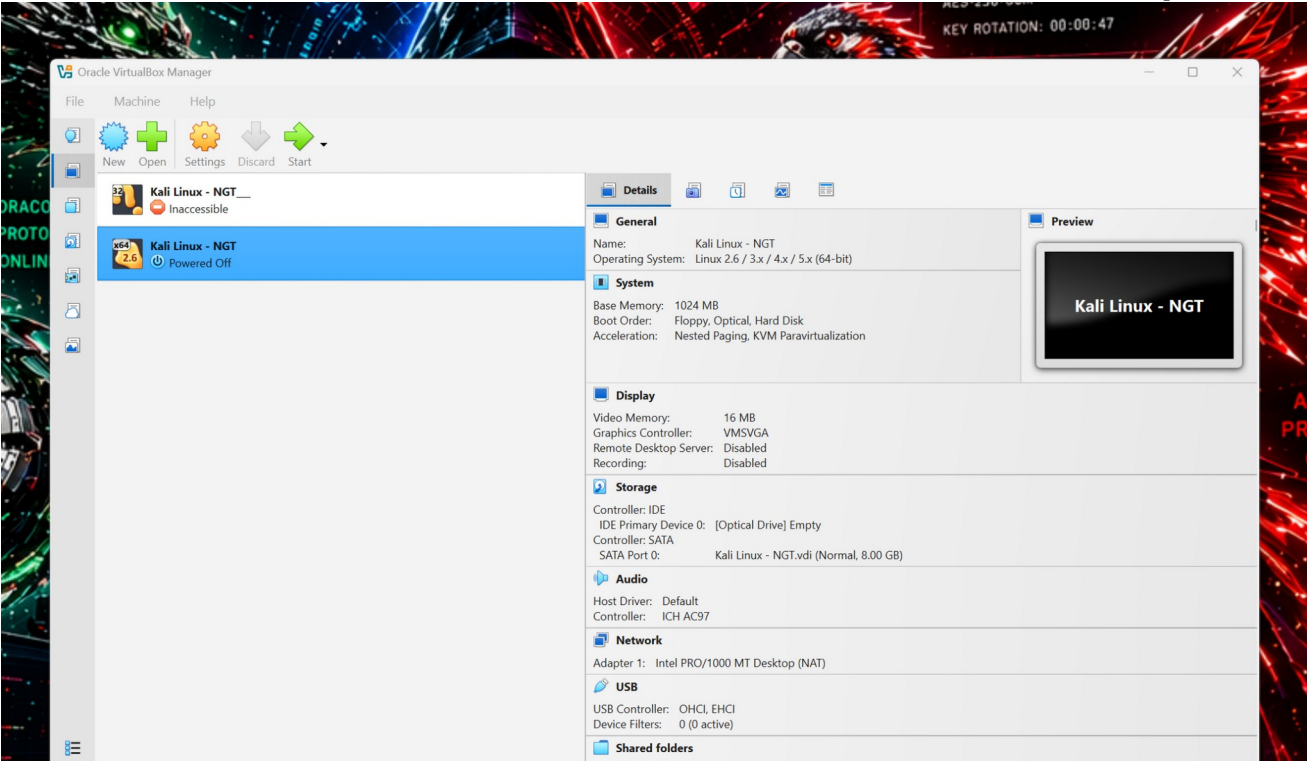


Figure 19. Final Kali VM configuration after remediation

Operational Procedures

- Shut down guests cleanly before moving or backing up VM folders.
- Keep the .vbox configuration and all referenced .vdi files together.
- Create snapshots before major updates, network changes, or tool installations.
- Use NAT or internal networks by default; avoid bridged mode unless a lab specifically requires it.
- Export important appliances to OVA and store a separate backup copy.
- Verify VirtualBox and Extension Pack versions remain matched after upgrades.

Interview Talking Points

- Installed and configured a desktop hypervisor on Windows 11 with elevated driver installation.
- Designed two isolated NAT lab networks with DHCP.
- Imported and tuned an enterprise-style Kali appliance with 8 GB RAM, 4 vCPUs, and an 80 GB disk.
- Resolved Extension Pack privilege errors and duplicate/misplaced VM registration issues.
- Validated virtual disks, configuration files, and guest network interfaces end to end.